

WORK SHEET - 01

VECTOR & 3D

[SINGLE CORRECT CHOICE TYPE]

- Q.1 If the distance of any point (x, y) from the origin is defined as $d(x, y) = \max\{|x|, |y|\}$ where $|\cdot|$ represents the absolute value function, and $d(x, y) = a$ where a is a non-zero constant, then the locus of the resulting curve is,
(A) Circle (B) Straight Line (C) Square (D) Triangle
- Q.2 The point on the line segment joining the points $P_1(1, 4, -3)$ and $P_2(1, 5, -1)$ which is $\frac{2}{3}$ of the way from P_1 to P_2 is
(A) $\left(1, -\frac{14}{3}, \frac{5}{3}\right)$ (B) $\left(-1, \frac{14}{3}, -\frac{5}{3}\right)$ (C) $\left(1, \frac{14}{3}, -\frac{5}{3}\right)$ (D) $\left(1, \frac{14}{3}, \frac{5}{3}\right)$
- Q.3 If the equation of plane perpendicular to the line $\frac{x-1}{2} = y-2 = \frac{z+1}{4}$ and containing the point $P = (-1, 3, 1)$ is $ax + y + bz - c = 0$ then the value of $a + b + c$ is
(A) 1 (B) 11 (C) 16 (D) None

[MULTIPLE CORRECT CHOICE TYPE]
(ONE OR MORE THAN ONE)

- Q.4 If $\sin(\alpha + \beta) \sin(\alpha - \beta) = \sin \gamma (2 \sin \beta + \sin \gamma)$, $0 < \alpha, \beta, \gamma < \pi$ then the family of lines $x \sin \alpha + y \sin \beta + \sin \gamma = 0$ passes through
(A) $(-1, 1)$ (B) $(1, 1)$ (C) $(1, -1)$ (D) $(-1, -1)$
- Q.5 The equation of a plane containing the point $P(-1, 2, 1)$, and is parallel to the line $l: x = 0, y = -z$ and its angle with line $l_1: x = y, z = 0$ is $\frac{\pi}{4}$ can be
(A) $x + 1 = 0$ (B) $x + 2y + 2z = 5$ (C) $y - 2 = 0$ (D) $2x + y + z = 1$
- Q.6 Consider the lines $l_1: \frac{x+y}{x+y+z} = 2$ and $l_2: x = y = z + 1$. The point(s) on l_1 which is a distance $\sqrt{\frac{14}{3}}$ from l_2 can be
(A) $(0, 2, -2)$ (B) $(2, 0, 2)$ (C) $(1, 1, 0)$ (D) $(3, -1, 4)$

[PARAGRAPH TYPE]

- Q.7 Let a point A is on the circle $x^2 + y^2 = r^2$. Two lines are drawn such that one of them is perpendicular to x-axis and intersect x-axis at point B, and a second line is perpendicular to y-axis and intersect the circle again at point C.
(i) The set of points where lines BC and AO (where O is origin) intersect is
(ii) Find r if the locus passes through $\left(\frac{3}{2}, -2\right)$.

[NUMERICAL INTEGER TYPE]

- Q.8 A straight line passing through A(6, 8) meets the curves $2x^2 + y^2 = 2$ at B and C. P is a point on BC such that the distance AB, AP and AC are in Harmonic Progression. Find minimum distance from origin to the locus of P.
- Q.9 A variable plane which remains at a constant distance $3p$ from the origin cuts the coordinate axes at A, B and C respectively. If that the locus of the centroid of ΔABC is $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{\lambda}{p^2}$, find λ .
- Q.10 Let ABC be a triangle with slopes of the sides AB, BC, CA are 2, 3, 5 respectively. Given origin is the orthocentre of the triangle ABC. Then find the locus of the centroid of the triangle ABC.
- Q.11 Let $a, b \in \mathbb{R}$. For a point $P = (x, y)$ let $f(P) = ax + by$. The line $ax + by = 4$ intersects the line segment joining two points P_1 and P_2 in the plane at a point R such that $P_1R : RP_2 = 1 : 2$. If $f(P_1) = 3$, then find the value of $f(P_2)$
- Q.12 Let $\vec{u}, \vec{v}, \vec{w}$ are three non-coplanar unit vectors and α, β, γ are the angles between $\vec{u}$ and $\vec{v}, \vec{v}$ and $\vec{w}, \vec{w}$ and $\vec{u}$ respectively and $\vec{x}, \vec{y}, \vec{z}$ are the unit vectors along the bisectors of the angles α, β, γ respectively. If $[\vec{x} \times \vec{y} \quad \vec{y} \times \vec{z} \quad \vec{z} \times \vec{x}] = \lambda [\vec{u} \quad \vec{v} \quad \vec{w}]^2 \sec^2 \frac{\alpha}{2} \sec^2 \frac{\beta}{2} \sec^2 \frac{\gamma}{2}$, find the value of $\frac{1}{\lambda}$.
- Q.13 Let $\vec{m}, \vec{n}, \vec{p}$ are 3 vectors such that $\vec{m} = (3t-1)\vec{i} + 2\vec{j} + 2\vec{k}$, $\vec{n} = (3-8t)\vec{i} + (t-2)\vec{j} - 3t\vec{k}$ and $\vec{p} = (2t-6)\vec{i} + 4\vec{j} - 3\vec{k}$. The value of parameter t so that the three vectors are linearly dependent is,
- Q.14 Find the equation of the plane containing the line of intersection of the planes $x + y + z = 0$ and $2x + y - 3z = 2$, and perpendicular to the plane $x - 2y - 5z = 17$.
- Q.15 Let T' and p' be the orthogonal projections of the point $T(-8, 2, -3)$ and the line $\ell : \frac{x}{4} = \frac{y-4}{3} = \frac{z+1}{-2}$ onto the plane $\pi : x - y + 3z + 8 = 0$. Find the distance between the point T' and the line p' .
- Q.16 The position vectors of the four angular points of a tetrahedron OABC are $(0, 0, 0); (0, 0, 2); (0, 4, 0)$ and $(6, 0, 0)$ respectively. A point P inside the tetrahedron is at the same distance r from the four plane faces of the tetrahedron. Find the value of r .
- Q.17 Let the plane $x + y = 0$ and two lines $l_1 : \frac{x}{3} = \frac{y+1}{1} = \frac{z-3}{-2}$ and l_2 (form with 2 planes): $y = z + 2$ and $x = 1$ find the line L that is parallel to the first plane and it intersects l_1 and l_2 in two points which is at a distance of 3 units.
- Q.18 If the line of intersection of the planes $x + 2y + 3z = 8$ and $2x + 3y + 4z = 11$ is coplanar with the line $\frac{x+1}{1} = \frac{y+1}{2} = \frac{z+1}{k}$, find k

ANSWER KEY

- Q.1 (C) Q.2 (C) Q.3 (B) Q.4 (A)
- Q.5 (AB) Q.6 (AB) Q.7 $x^2 + y^2 = \frac{r^2}{9}$ and 7.5
- Q.8 $\frac{1}{\sqrt{52}}$ Q.9 [1] Q.11 [6] Q.12 [16]
- Q.13 $4, \frac{3}{11}$ Q.14 $9x + 7y - z = 4$ Q.15 $\frac{2}{\sqrt{6}}$ Q.16 $\frac{2}{3}$
- Q.17 $\vec{r} = (1, 2, 0) + \lambda (2, -2, 1)$ or $\vec{r} = \left(1, \frac{10}{9}, \frac{-8}{9}\right) + \lambda (12, -12, 21)$
- Q.18 [3]

Rankers Academy JEE

